

BS (Data Science) Part-III
DASC- 523 Parallel and Distributed Computing

LAB # 3 – Mutual exclusion in OpenMP

Lab Objective: To understand and work with the thread synchronization in OpenMP for multithreaded programming.

Terminology:

- Mutual exclusion: Only one thread at a time can enter a critical region.
- Critical and Atomic constructs of OpenMP provide mutual exclusion. Atomic clause provides mutual exclusion but only applies to the update of a memory location.
- `#pragma omp critical` – for a code block (C/C++)
- `#pragma omp atomic` – for single statements

Task - Write a synchronized multithreaded program providing mutual exclusion using critical clause.

```
#include <omp.h>
#include <stdio.h>
int main() {
    omp_set_num_threads(10);
    int i = 1;
    #pragma omp parallel
    {
        #pragma omp critical
        {
            int tid = omp_get_thread_num();
            printf("Value of %d from thread %d \n", i, tid);
            i = i+1;
        }
    }
    return 0;
}
```

Submitted
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18/9/20